

APO-CARVEDILOL TABLETS USP

6.25 MG & 25 MG

Congestive Heart Failure Agent

Pharmacology

Carvedilol is a cardiovascular agent for the treatment of congestive heart failure that combines beta-adrenoreceptor blockade and vasodilation in a single racemic mixture. Nonselective beta-adrenoreceptor blocking activity is present in the S(-) enantiomer and α_1 -adrenoreceptor blocking activity is present at equal potency in both the R(+) and S(-) enantiomers. Carvedilol has no intrinsic sympathomimetic activity. Its action on beta-receptors is 10 times stronger than on α -receptors.

Carvedilol reduces peripheral vascular resistance by vasodilation, thereby causing a fall in systemic blood pressure after acute administration, predominantly mediated through selective α_1 -antagonism. Beta blockade prevents reflex tachycardia with the net result that heart rate is unchanged or decreased. Carvedilol reduces renin release through beta blockade.

Pharmacokinetics:

Carvedilol is rapidly absorbed following oral administration, with peak plasma concentrations of carvedilol observed at 1 hour post-dose in fasting subjects. Despite being well absorbed, absolute bioavailability is approximately 25% to 35% due to a significant degree of first-pass metabolism.

Carvedilol is highly bound to plasma proteins, (greater than 98%) primarily to albumin. The plasma-protein binding is independent of concentration over the therapeutic range. Carvedilol is a basic lipophilic compound with a steady-state volume of distribution of approximately 115 L.

Following oral administration, the apparent mean terminal elimination half-life of carvedilol ranges from 7 to 10 hours. Plasma clearance ranges from 500 to 700 mL/min. Carvedilol is extensively metabolized with less than 2% of the dose excreted unchanged in the urine. Carvedilol is metabolized mainly by glucuronidation and aromatic ring oxidation by the cytochrome P450 system (primarily CYP2D6 and CYP2C9 isozymes). The metabolites of carvedilol are excreted mainly via the bile into the feces. Elimination is mainly biliary. The primary route of excretion is via the feces. A minor part is eliminated via the kidneys in the form of various metabolites.

Indication

Hypertension: APO-CARVEDILOL (carvedilol) is indicated primarily for the management of essential hypertension. It can be used alone or in combination with other antihypertensive agents (e.g. calcium channel blockers, diuretics).

Treatment of angina pectoris.

Treatment of symptomatic chronic heart failure (CHF): APO-CARVEDILOL (carvedilol) is indicated for the treatment of symptomatic CHF to reduce mortality and cardiovascular hospitalizations, improve patient well-being and slow the progression of the disease. APO-CARVEDILOL may be used as adjunct of standard therapy, but may also be used in those patients unable to tolerate an ACE inhibitor, or those who are not receiving digitalis, hydralazine or nitrate therapy.

Recommended Dose and Administration

Duration of treatment: Treatment with carvedilol is a long-term therapy. Treatment should not be stopped abruptly but

rather gradually reduced at weekly intervals. This is particularly important in the case of patients with concomitant coronary heart disease.

Essential hypertension: The recommended dose for initiation of therapy is 12.5 mg once a day for the first 2 days. Thereafter, the recommended dosage is 25 mg once a day. If necessary, the dosage may subsequently be increased at intervals of at least two weeks to the recommended maximum daily dose of 50 mg given once daily or in divided doses (twice daily).

Angina pectoris: The recommended dose for initiation of therapy is 12.5 mg twice a day for the first 2 days. Thereafter the recommended dosage is 25 mg twice a day. If necessary, the dosage may thereafter be increased at intervals of at least two weeks, up to the recommended maximum daily dose of 100 mg given in divided doses (twice daily).

Symptomatic, stable, chronic heart failure: Dosage must be tailored to suit the individual and closely monitored by a physician during up-titration. For those patients receiving digitalis, diuretics and ACE inhibitors, dosing of these drugs should be stabilized before initiation of carvedilol treatment. The recommended dose for initiation of therapy is 3.125 mg twice daily for 2 weeks. If this dose is tolerated, the dose may thereafter be increased, at intervals of not less than two weeks, to 6.25 mg, 12.5 mg and 25 mg twice daily. Doses should be increased to the highest level tolerated by the patient.

The maximum recommended dose is 25 mg twice daily for all patients with severe CHF and for patients with mild to moderate CHF weighing less than 85 kg (187 lbs).

In patients with mild or moderate CHF weighing more than 85 kg, the maximum recommended dose is 50 mg twice daily. Before each dose increase, the patient should be evaluated by the physician for symptoms of vasodilation or worsening heart failure. Transient worsening of heart failure or fluid retention should be treated with increased doses of diuretics. Occasionally, it may be necessary to lower the dose of carvedilol and, in rare cases, temporarily discontinue carvedilol treatment. If Carvedilol treatment is discontinued for more than one week, therapy should be recommenced at a lower dose level (twice daily) and up-titrated in line with the above dosing recommendation. If carvedilol is discontinued for more than two weeks, therapy should be recommenced at 3.125 mg in line with the above dosing recommendation. Symptoms of vasodilation may be managed initially by a reduction in the dose of diuretics. If symptoms persist, the dose of ACE inhibitor (if used) may be reduced, followed by a reduction in the dose of carvedilol if necessary. Under these circumstances, the dose of carvedilol should not be increased until symptoms of worsening heart failure or vasodilation have been stabilized.

Special dosage instructions:

Renal impairment: Available pharmacokinetic data in patients with varying degrees of renal impairment (including renal failure) suggest no changes in carvedilol dosing recommendations are warranted in patients with moderate to severe renal insufficiency.

Hepatic impairment: Carvedilol is contraindicated in patients with clinical manifestations of liver dysfunction (see Contraindications). Elderly There is no evidence to support dose adjustment.

Method of administration: The tablets are to be swallowed with sufficient fluid.

Contraindication

Carvedilol is contra-indicated in patients with marked fluid retention or overload requiring intravenous inotropic support.

Patients with obstructive airways disease, liver dysfunction, hypersensitivity to carvedilol or any other constituent of the tablets.

As with other beta-blocking agents: History of bronchospasm or asthma, 2nd and 3rd degree A-V heart block, (unless a permanent pacemaker is in place), severe bradycardia (< 50 bpm), cardiogenic shock, sick sinus syndrome (including sino-atrial block), severe hypotension (systolic blood pressure < 85mmHg), metabolic acidosis and phaeochromocytoma (unless adequately controlled by alpha blockade).

Warnings

In chronic heart failure patients, worsening cardiac failure or fluid retention may occur during up-titration of carvedilol. If such symptoms occur, the dose of diuretic should be adjusted and the carvedilol dose should not be advanced until clinical stability resumes. Occasionally it may be necessary to lower the carvedilol dose or temporarily discontinue it. Such episodes do not preclude subsequent successful titration of carvedilol.

In hypertensive patients who have chronic heart failure controlled with digoxin, diuretics and/or an ACE inhibitor, carvedilol should be used with caution since both digoxin and carvedilol may slow A-V conduction.

As with other drugs with beta-blocking activity, carvedilol may mask the early signs of acute hypoglycaemia in patients with diabetes mellitus. Alternatives to beta-blocking agents are generally preferred in insulin-dependent patients. In patients with diabetes, the use of carvedilol may be associated with worsening control of blood glucose. Therefore, regular monitoring of blood glucose is required in diabetics when carvedilol is initiated or up-titrated and hypoglycaemic therapy adjusted accordingly.

Reversible deterioration of renal function has been observed with carvedilol therapy in chronic heart failure patients with low blood pressure (systolic BP < 100mmHg), ischaemic heart disease and diffuse vascular disease, and/or underlying renal insufficiency. In CHF patients with these risk factors, renal function should be monitored during up-titration of carvedilol and the drug discontinued or dosage reduced if worsening of renal failure occurs.

Wearers of contact lenses should be advised of the possibility of reduced lacrimation.

Although angina has not been reported on stopping treatment, discontinuation should be gradual (1 - 2 weeks) particularly in patients with ischaemic heart disease, as carvedilol has beta-blocking activity.

Carvedilol may be used in patients with peripheral vascular disease. Pure beta-blockers can precipitate or aggravate symptoms of arterial insufficiency. However as carvedilol also has alpha-blocking properties this effect is largely counterbalanced.

Carvedilol, as with other agents with beta-blocking activity, may mask the symptoms of thyrotoxicosis.

If carvedilol induces bradycardia, with a decrease in pulse rate to less than 55 beats per minute, the dosage of carvedilol should be reduced.

Care should be taken in administering carvedilol to patients with a history of serious hypersensitivity reactions and in those undergoing desensitisation therapy as beta-blockers may increase both the sensitivity towards allergens and the seriousness of anaphylactic reactions.

In patients suffering from the peripheral circulatory disorder Raynaud's phenomenon, there may be exacerbation of symptoms.

Patients with a history of psoriasis associated with beta-blocker therapy should be given carvedilol only after consideration of the risk-benefit ratio.

In patients with phaeochromocytoma, an alpha-blocking agent should be initiated prior to the use of any beta-blocking agent. There is no experience of the use of carvedilol in this condition. Therefore, caution should be taken in the administration of carvedilol to patients suspected of having phaeochromocytoma.

Agents with non-selective beta-blocking activity may provoke chest pain in patients with Prinzmetal's variant angina. There is no clinical experience with carvedilol in these patients, although the alpha-blocking activity of carvedilol may prevent such symptoms. However, caution should be taken in the administration of carvedilol to patients suspected of having Prinzmetal's variant angina.

In patients with a tendency to bronchospastic reactions, respiratory distress can occur as a result of a possible increase in airway resistance.

Drug Interaction

As with other agents with beta-blocking activity, carvedilol may potentiate the effect of other concomitantly administered drugs that are anti-hypertensive in action (e.g. alpha₁-receptor antagonists) or have hypotension as part of their adverse effect profile.

Patients taking an agent with β-blocking properties and a drug that can deplete catecholamines (e.g. reserpine and monoamine oxidase inhibitors) should be observed closely for signs of hypotension and/or severe bradycardia.

Isolated cases of conduction disturbance (rarely with haemodynamic disruption) have been observed when carvedilol and diltiazem were given concomitantly. Therefore, as with other drugs with beta-blocking activity, careful monitoring of ECG and blood pressure should be undertaken when co-administering calcium channel blockers of the verapamil or diltiazem type, or class I antiarrhythmic drugs. These types of drugs should not be co-administered intravenously in patients receiving carvedilol.

The effects of insulin or oral hypoglycaemics may be intensified. Regular monitoring of blood glucose is therefore recommended.

Trough plasma digoxin levels may be increased by approximately 16% in hypertensive patients co-administered carvedilol and digoxin. Increased monitoring of digoxin levels is recommended when initiating, adjusting or discontinuing carvedilol. Concomitant administration of carvedilol and cardiac glycosides may prolong AV conduction time.

When treatment with carvedilol and clonidine together is to be terminated, carvedilol should be withdrawn first, several days before gradually decreasing the dosage of clonidine.

Care may be required in those receiving inducers of mixed function oxidases e.g. rifampicin, as serum levels of carvedilol may be reduced or inhibitors of mixed function oxidases e.g. cimetidine, as serum levels may be increased. During general anaesthesia, attention should be paid to the potential synergistic negative inotropic effects of carvedilol and anaesthetic drugs.

Modest increases in mean trough cyclosporin concentrations were observed following initiation of carvedilol treatment in 21 renal transplant patients suffering from chronic vascular rejection. In about 30% of patients, the dose of cyclosporin had to be reduced in order to maintain cyclosporin concentrations within the therapeutic range, while in the remainder no adjustment was needed. On average, the dose of cyclosporin was reduced about 20% in these patients. Due to wide interindividual variability in the dose adjustment required, it is recommended that cyclosporin concentrations be monitored closely after initiation of carvedilol therapy and that the dose of cyclosporin be adjusted as appropriate.

Use in pregnancy and lactation

There is no adequate experience with carvedilol in pregnant women.

Carvedilol should not be used in pregnancy unless the anticipated benefits outweigh the potential risks.

Animal studies have shown that carvedilol or its metabolites are excreted in breast milk. It is not known whether carvedilol is excreted in human milk. Breast feeding is therefore not recommended during the administration of carvedilol.

Adverse Effects

Adverse events are listed separately for CHF because of differences in the background diseases.

In chronic heart failure

Haematological

Rare, thrombocytopenia.

Leucopenia has been reported in isolated cases.

Metabolic

Commonly: weight increase and hypercholesterolaemia. Hyperglycaemia, hypoglycaemia and worsening control of blood glucose are also common in patients with pre-existing diabetes mellitus.

Central nervous system

Very common: dizziness, headache are usually mild and occur particularly at the start of treatment. Asthenia (including fatigue).

Cardiovascular system

Common: bradycardia, postural hypotension, hypotension, oedema (including generalised, peripheral, dependent and genital oedema, oedema of the legs, hypervolaemia and fluid overload).

Uncommon: syncope (including presyncope), AV-block and cardiac failure during up-titration.

Gastro-intestinal system

Commonly: nausea, diarrhoea, and vomiting.

Skin and appendages

Dermatitis, and increased sweating.

Others

Commonly: vision abnormalities.

Rarely: acute renal failure and renal function abnormalities in patients with diffuse vascular disease and/or impaired renal function.

The frequency of adverse experiences is not dose dependent, with the exception of dizziness, abnormal vision and bradycardia.

In hypertension and angina:

The profile is similar to that observed in chronic heart failure although the incidence of events is generally lower in patients with hypertension or angina treated with carvedilol.

Blood chemistry and haematological

Isolated cases of changes in serum transaminases, thrombocytopenia and leucopenia have been reported.

Central nervous system

Commonly: dizziness, headaches and fatigue, which are usually mild and occur particularly at the beginning of treatment.

Uncommonly: depressed mood, sleep disturbance, paraesthesia, asthenia.

Metabolic

Due to the beta-blocking properties it is also possible for latent diabetes mellitus to become manifest, manifest diabetes to be aggravated, and blood glucose counter-regulation to be inhibited.

Cardiovascular system

Commonly: bradycardia, postural hypotension, especially at the beginning of treatment.

Uncommonly, syncope, hypotension, disturbances of peripheral circulation (cold extremities, PVD, exacerbation of intermittent claudication and Raynauds phenomenon). AV-block, angina pectoris (including chest pain), symptoms of heart failure and peripheral oedema.

Respiratory system

Commonly: asthma and dyspnoea in predisposed patients.

Rarely: stuffy nose. Wheezing and flu-like symptoms.

Gastro-intestinal system

Commonly: gastro-intestinal upset (with symptoms such as nausea, abdominal pain, diarrhoea).

Uncommonly: constipation and vomiting.

Skin and appendages

Uncommonly: skin reactions (e.g. allergic exanthema, dermatitis, urticaria, pruritus, lichen planus-like reactions, and increased sweating). Psoriatic skin lesions may occur or existing lesions exacerbated.

Others

Commonly: pain in the extremities, reduced lacrimation.

Uncommon: cases of sexual impotence and disturbed vision.

Rarely: dryness of the mouth and disturbances of micturition and eye irritation.

Isolated cases of allergic reactions have been reported.

Symptoms and Treatment of Overdose

Symptoms and signs

Profound cardiovascular effects such as hypotension and bradycardia would be expected after massive overdose. Heart failure, cardiogenic shock and cardiac arrest may follow. There may also be respiratory problems, bronchospasm, vomiting, disturbed consciousness and generalised seizures.

Treatment

Gastric lavage or induced emesis may be useful in the first few hours after ingestion.

In addition to general procedures, vital signs must be monitored and corrected, if necessary under intensive care conditions.

Patients should be placed in the supine position. Atropine, 0.5mg to 2mg i.v. and/or glucagon 1 to 10mg i.v. (followed by a slow i.v. infusion of 2 to

5mg/hour if necessary) may be given when bradycardia is present. Pacemaker therapy may be necessary. For excessive hypotension, intravenous fluids may be administered. In addition, norepinephrine may be given, either 5 to 10 micrograms i.v., repeated according to blood pressure response, or 5 micrograms per minute by infusion titrated to blood pressure. Bronchospasm may be treated using salbutamol or other beta₂-agonists given as aerosol or, if necessary, by the intravenous route. In the event of seizures, slow i.v. injection of diazepam or clonazepam is recommended.

In cases of severe overdose with symptoms of shock, supportive treatment as described should be continued for a sufficiently long period of time, i.e. until the patient stabilises, since prolonged elimination half life and redistribution of carvedilol from deeper compartments can be expected.

Storage Conditions

Store below 30°C in a dry place. Protect from high humidity and light.

Availability of Dosage Forms

6.25mg: Each oval, white, film-coated tablet engraved 'APO' on one side and '6.25' on the other contains 6.25 mg carvedilol. Available in bottle of 30 tablets and 100 tablets. Not all pack sizes are marketed.

25mg: Each oval, white, film-coated tablet engraved 'APO' on one side and 'C25' on the other contains 25 mg carvedilol. Available in bottle of 30 tablets and 100 tablets. Not all pack sizes are marketed.

Shelf-life

24 months from the date of manufacture.

Manufacturer

Apotex Inc., 150 Signet Dr, Toronto (Weston), Ontario, Canada M9L 1T9.

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Apotex Inc.

150 Signet Drive, Weston
Ontario, Canada M9L 1T9

Distributor : Pharmaforte (M) Sdn Bhd

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